Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/19/2023
PI Name:	Martin, Bryn Ph.D.		
Project Title:	Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	83843	**Congressional District:**	1
Comments:	PI moved to Alcyone Therapeutics, Inc., in late 2020 from University of Idaho (Ed., December 2021) PI moved to Neela Therapeutics in late 2025 from Alcyone (Ed, November 2025)		
Project Type:	Ground	**Solicitation / Funding Source:**	2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C
Start Date:	12/08/2021	**End Date:**	12/07/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date of new grant 80NSSC22K0023 is 12/07/2024 per grant information documents (Ed., December 2021). NOTE: End date of previous grant number 80NSSC20K0920 changed to 10/12/2020 per JSC Grants Admin/NSSC information, due to PI move and new grant established (Ed., 12/10/21).		
Key Personnel Changes/Previous PI:	There have not been any key personnel changes in the last reporting period.		
COI Name (Institution):	Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Macias, Brandon Ph.D. (NASA Johnson Space Center) Marshall-Goebel, Karina Ph.D. (KBR/NASA Johnson Space Center) Williams, Michael M.D. (University of Washington, Seattle) Loerch, Linda Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K0023		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NOTE: Continuation of "Ophthalmic and Intracranial Structural Changes in Head-Down Tilt Bedrest: Potential Countermeasures and Comparison to SANS Findings in Astronauts (OPTICS study)," grant 80NSSC20K0920, with the same Principal Investigator (PI) Dr. Bryn Martin. The proposed research plan builds on our group's ongoing research to quantify ophthalmic and intracranial changes in long duration spaceflight astronauts using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. We propose to use identical techniques currently implemented pre- and post-spaceflight to also quantify how long-duration bed rest and countermeasures applied during bed rest impact ophthalmic and intracranial structures. It is critically essential to use identical techniques that we have already applied in astronauts to determine if structural alterations seen in bed rest are similar to what our team has already quantified in astronauts with spaceflight associated neuro-ocular syndrome (SANS). Similarly, it is essential to use the identical techniques to quantify the capability of a countermeasure to mitigate structural alterations in bed rest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help better understand ophthalmic and cerebrospinal fluid physiology and anatomy that can potentially be used for diagnostic and therapeutic applications of terrestrial human diseases such as hydrocephalus and cerebrospinal fluid drug delivery.
Task Progress:	Spaceflight has been hypothesized to result in a bulk body fluid shift toward the head. The OPTICS study is conducted on a group of healthy subjects undergoing a head-down tilt (HDT) procedure in an attempt to mimic fluid shifts observed in space and implement potential countermeasures to evaluate their potential efficacy in mitigating the negative effects of long-duration spaceflight (LDSF) on sensitive eye and brain tissues due to fluid shifts. This research builds on our group's ongoing research to quantify ophthalmic and intracranial changes in long-duration spaceflight in astronauts using novel non-invasive magnetic resonance imaging (MRI) protocols and post-processing techniques. Key progress in the 2023 fiscal year involved completing subject recruitment and measure collection, as well as continuing data analysis. Forty-eight subjects (age range 24:54) were recruited and enrolled in study protocols for head-down tilt at the German Aerospace Center Institute for Aerospace Medicine (DLR) in Cologne, Germany. Subjects were screened such that all were age and gender matched across each group, with the exclusion criteria of: past or present ocular pathology, presence of connective tissue disorder, spinal fusion, previous history of neurological and/or cardiovascular disease and/or surgery, intracranial mass/deformity, cerebrospinal fluid (CSF) leak, spinal cord tethering, spina bifida, myelomeningocele, and implanted metal devices. Prior to each magnetic resonance (MR) scan, the subject height, weight, waist circumference, heart rate, blood pressure, and other biometrics were recorded. For campaign 1 and 2 (2022 year), participants were divided equally into two groups of interventions: * Group 4) lower body negative pressure (LBNP) for 3-hour sessions twice daily, and * Group 1) seated position for 3-hour sessions twice daily. Interventions were divided into morning and afternoon sessions for both groups. The LBNP intervention was maintained at 25 mmHg ? 2 mmHg. Campaigns 3 and 4 were completed during the 2023 year, also divided into two groups: One group being the daily exercise and thigh cuff group (Group 3), and another the bedrest control group (Group 2). There were no significant changes between optic globe volume displacement from baseline in the seated vs. lower body negative pressure groups at day 15 or day 29. There was a trend upward in mean globe volume displacement from combined groups on day 15 to day 29, although it was not significant. Large volume displacement in either countermeasure group was rare, with only two subjects (three eyes) displaying a change greater than 10 mm3, both in the lower body negative pressure group. The potential impact of each countermeasure will be quantified with respect to the control group. Final conclusions are pending control and exercise group data analysis for the study in 2024. MRI data for the investigation of the change of brain and CSF volumes has been collected and examined for image quality; data analyses have begun and initial results for volumetric changes have been calculated. Brain and CSF results are to be presented by Dr. Kramer et al. at the upcoming 2024 NASA Human Research Program (HRP) conference. Dr. Warthen is currently evaluating brain shift for this study in terms of the individual brain shift within the skull, before and after head-down tilt. This data will be compared to the astronaut and control cohort, also to be presented at the upcoming 2024 HRP conference. Going forward, we plan to finish the primary aims' data analysis, along with comparing all groups with astronauts and controls.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Sater SH, Sass AM, Seiner A, Natividad GC, Shrestha D, Fu AQ, Oshinski JN, Ethier CR, Martin BA. "MRI-based quantification of ophthalmic changes in healthy volunteers during acute 15° head-down tilt as an analogue to microgravity." J R Soc Interface. 2021 Apr;18(177):20200920. https://doi.org/10.1098/rsif.2020.0920 ; PMID: 33906382; PMCID: PMC8086909 , Apr-2021